

xMCF

ISO/PAS 8329

Description of mechanical connections
and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
focus group on issues [#95](#) / [#104](#)
2026-10-07

Meeting results are
added in green font.
Additions are
inserted in light blue.

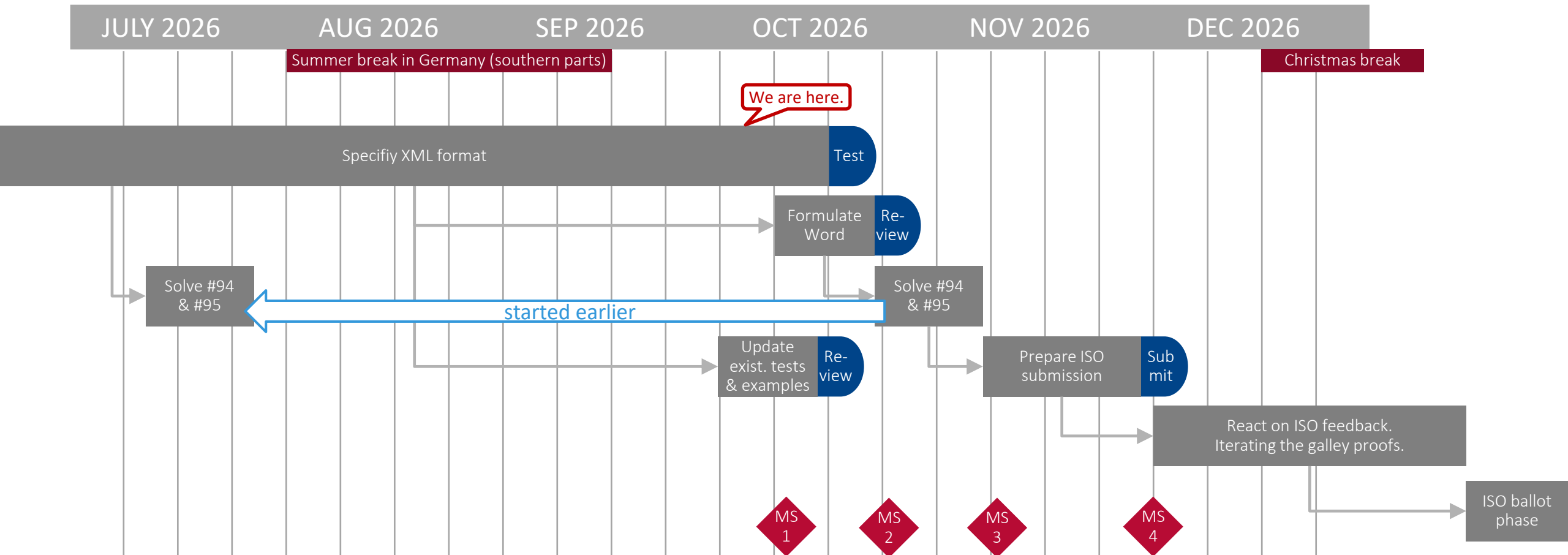
Agenda

As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-10-02_VideoConference/README.md

- Welcome & Flying over today's topics
- Timeline for next release
- Question from PTC: Multiple `<loc_list/>` vs. continuous `<weld_position/>` propagation.
— according GitHub issues have been created.
- Cooperation Model
- Work in progress — focus on issues [#95](#) / [#94](#) / [#106](#) / [#105](#) / [#61](#), as discussed last time.
- Focus on XML / DOC suggestions, XSD- / Schematron-definition & examples
- Steps we should take next



Timeline for the next release



Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.

Multiple <loc_list/> vs. continuous <weld_position/> propagation.

From example “chapter_10_2_5_4_exampleB.xml”:

```
<connection_1d>
  <loc_list index="1"> <!-- first section -->
a      <loc v="0"    > 2581.21 -708.408 31.6532 </loc>
b      <loc v="1"    > 2581.42 -708.357 35.2816 </loc>
c      <loc v="2.22"> 2581.05 -708.302 39.0643 </loc>
  </loc_list>
  <loc_list index="2"> <!-- second section -->
d      <loc v="1"    > 2581.05 -708.302 40.3340 </loc>
e      <loc v="2.1"> 2581.05 -708.302 48.5300 </loc>
  </loc_list>

  <seamweld>
    <butt_joint base="1">
      <weld_position u="0.2" x="1" y="0" z="1"/>
    </butt_joint>
  </seamweld>
</connection_1d>
```

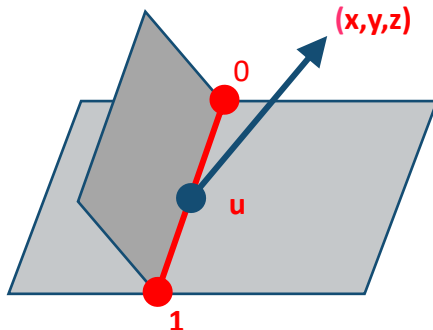


Figure 52 — Welding position of a Y-joint

From clause 10.2.4.4.4:

The attribute *u* specifies the relative location on the <loc_list/> polyline. Value *u*=0 represents the first location of the connection line matching the element <loc/> specified with the lowest value for the attribute *u*. Value *u*=1 represents the last location of this line matching the element <loc/> with highest value for the attribute value *u*. Values in between are specifying the point located at the specified fraction of the <loc_list/> polyline measure in summed up lengths of the segments of the <loc_list/> polyline in space.

This does *not* tell us how to handle *more than one* <loc_list/>.

Q1: The foot point of <weld-position/>-direction is

$$a + u * (|b - a| + |c - b|) * \frac{b - a}{|b - a|} = (2581.296, -708.387, 33.138)$$

which is roughly in the middle between *a* and *b*.

— If we consider the first section, only! But what about *|e - d|*?

Q2: The <weld-position/>-direction should be maintained continuously along the weld line. But what if the weld line is interrupted?

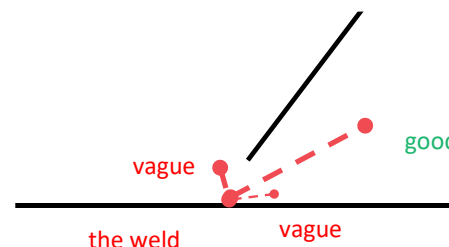
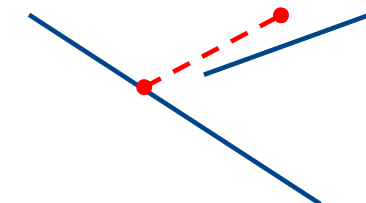


Figure 53 — Welding position vector direction and length

What if the situation at the 2nd <loc_list/> is like this?



Multiple <loc_list/>s vs. only one <weld_position/>

- Decision: We treat the set of all polylines as if it was continuous. Then, <weld_position/> can be propagated from one polyline to the next one. And for the definition of parameter “u”, all polyline lengths are summed up. ⇒ This answers Q2 & Q1.
- Decision criterion: If the robot *cannot* weld a line in one run, it must be split in two <connection_1d/> (and not in two <loc_list/>).
- This implies that gaps between adjacent <loc_list/> must not be too big.
- Recommendation for max. gap size: 10 × geometric tolerance of the software system. Here, “geometric tolerance” refers to the maximum distance between two coordinates for them to be considered numerically coincident.
- The same applies to <adhesive_line/>, although gaps do not play such a significant role when applying adhesive, as the robot does not need to start again.
- In consequence, the formula from Q1 must look like this:
$$a + u * (|b - a| + |c - b| + |e - d|) * \frac{b-a}{|b-a|},$$
where u is so small that the foot point lies within <loc_list index="1">
- This decision needs to be documented in the specification.
The [issue #117](#) has been created for this.

Explain multiple `<loc_list/>` vs. intermittent conn-lines

Would it be a good idea to add an explanation in case intermittent conn-lines are used in conjunction with multiple `<loc_list/>`?

It appears at least not to be trivial how the `from` and `to` parameters are to be handled in the vicinity of gaps ...

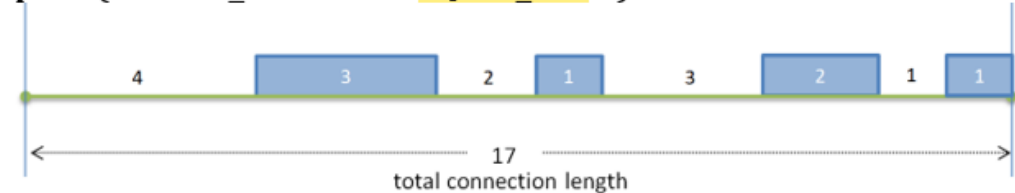
The suggestion “add explanation” was confirmed, e. g. in the following sense:

The strategy to handle multiple polylines as “one continuous geometrical entity” applies here, too.

Thus, the parameters “`from`” and “`to`” refer to the total length of all polylines. (The final wording has yet to be found.)

[The issue #118](#) has been created for adding this explanation.

Example D (a `<corner_weld/>` with `<segment_list/>`):



```
<seamweld>
  <corner_weld base="1" technology="resistance">
    <weld_position u="0.2" x="1" y="0" z="1">
      <segment_list>
        <segment from="4.0" to="7.0" />
        <segment from="9.0" to="10.0" />
        <segment from="13.0" to="15.0" />
        <segment from="16.0" to="17.0" />
      </segment_list>
    </weld_position>
    <sheet_parameter ... />
  </corner_weld>
</seamweld>
```

Issue #95: Add persistent identifier to `connection_group`

as of 2026-07-01
≙ 2026-07-08

Not discussed
2026-10-02.

- Original request: „Up to now, `connection_groups` have an integer identifier which is local to each `χMCF` file, and not persistent.
Under PDM considerations, an optional persistent alphanumeric id would help to increase the process reliability ...”
- This use case has been very well justified.
- However, there are other use cases of great practical relevance where the introduction of a new mandatory attribute requiring attention would result in a significant additional workload, without this being offset by any benefit.
- The group decided to **continue the discussion** at the **next meeting** with the question:
‘What if the attribute (persistent identifier) were just *optional*?’ → See next slide.
- The following important point was also raised during the discussion:
<connected_to/> lists the connected parts in *no particular order*.
The order may change even after a simple im- / export sequence.
If the stacking sequence is important, always use the <stacking/> element!
→ 2026-09-06: An according [issue #112](#) was created.



Issue #95: Add persistent identifier to `connection_group`

(continued) ‘What if the attribute (persistent identifier) were just *optional*?’

Results of
2026-07-22
& 2026-07-28

Not discussed
2026-10-02.

- PDM Use Cases: Exchange with Partner and recognize a known connection group upon re-import.
- CAD Use Cases: Creating, Deleting, Splitting, Merging, Editing of connections, incl. changing the connected parts.
- PEP is “owner” of the unique connection group IDs, just as with the part codes.
(Single Point of Truth to guarantee uniqueness!)
PDM system, as “incarnation” of the PEP, is the tool which creates and keeps authority on such IDs.
- Difference between part codes and connection group IDs:
For CAD data, there is (usually) only *one* CAD authoring system, which is tightly integrated with PDM.
For connection data, this frequently is different:
Many tools are established for manipulating connections, which are not tightly PDM-integrated.
- Solution ideas:
 - a. PDM system offers and tool attaches to a standardized interface for querying suitable connection group ID.
 - b. Tools could exchange one xMCF file per connection group
(which renders the connection group ID as meaningless, as it is currently.)
 - c. More ideas? --- no → Concept is confirmed (2026-09-16). → Text for spec update to be drafted in comments to issue #95.
- Integer connection group *id* shall be supplemented with an alphanumeric *label*.
2026-09-09: ⇒ XSD to be updated. / 2026-09-16: SH added according change to GitHub, this morning.

Cooperation Model

Next steps in expanding the standard text and how we coordinate our joint work on this

- CF to approach KLT for contributing some general intro about Schematron and a reference to STEP AP242. (Done)
- We need a new general clause which describes methods & means for automated validation, like Java-based XSD- and Schematron validators.
- CF to extend distribution script by Schematron file, such that it appears next to the [XSD on the ISO download server](#).
- For issues [#61](#) and [#105](#), there is a shared solution via “Templates”, so they should be worked on together.
- Work on issue [#95](#) can proceed independently of this. Same with [#94](#).
- In general, we need for each of these
 - Some words for the change log
 - New / changed clause(s) with explanation & motivation of the use case(s) and the solution concept.
 - Description and example files for the syntax
 - test cases / test files for the syntax
- “Small” issues [#112](#) ... [#118](#) can be picked by anyone, any time.

Modus operandi:

- The one who starts working on a binary file (especially the Word file) places a *semaphore* file "\${filename}_locked_by_XYZ.txt" next to the binary in GitHub, such that we can synchronize changes and avoid later merge conflicts.
- Let's focus on content, first, not on formatting. Especially page breaking is postponed.
- We can have separate branches for [#94](#) and [#95](#) (if necessary).



Issue #94: Add instance identifier to a connector

as of 2026-09-09

Not discussed
2026-10-02.

- Original request: „In order to keep consistency between the PDM system and the connection data, it is required to have an instance id for the rivets“ (Rivets or any other connection method in which a separate physical component establishes the connection.)
- The standard documents says in clause “8.2.2 Attribute `label`”: “Any connection should have an attribute called `label`, which identifies it throughout the entire CAX process, maybe even throughout the complete product lifecycle, including manufacturing. It is not necessary that these labels are unique.”
Remark 1: The use cases in which “`label`” is not unique are quite specific and rare.
Remark 2: “`label`” must not be confused with “`part_code`”, which frequently is *not* unique. Example:

```
<connection_0d label="BOLT_135" part_code="BOLT-NUT-Combi #17"> ... <!-- only label is unique -->
  <threaded_connection ... part_code="M10x50 8.8"/> <!-- not unique -->
</connection_0d>
<connection_0d label="BOLT_136" part_code="BOLT-NUT-Combi #17"> ... <!-- only label is unique -->
  <threaded_connection ... part_code="M10x50 8.8"/> <!-- not unique -->
</connection_0d>
```

In PDM, e.g. `label="BOLT_135"` would denote an *instance* combination of e.g. a bolt, some washers and a nut.

However, a part code *for this combination* is needed and not yet supported by xMCF.

⇒ Suitable place for the `part_code` for the combination to be found in xMCF syntax – at both, the actual connection plus its template, if there is one.

Suggestion: Put the `part_code` as shown above in blue, at both positions.

— **No**, it should be an attribute of `<threaded_connection/>`, see next slide!

2026-09-09: For discussion & validation, see following slides!



#94 wrt. templates

as of 2026-09-09

Not discussed
2026-10-02.

- The combination's `part_code` should *not* reside on `<connection_0d/>`, because there it would also be accessible for e.g. spotwelds. Instead, it should be an attribute of `<threaded_connection/>`. Thus, another name is needed for the combination's `part_code`, namely "`combination_part_code`".
- In case a connection template is used, the `part_code` of a connection may be given there.
- The minimal content of a `<connection_0d/>` would then be:
 - Label (highly recommended)
 - Template (as mentioned above)
 - `<loc/>`
- The part code of the connection (e.g. the bolt) and its components (screw, nut, washers) are then given only once, i. e. in the template definition.
- Remark: In any case, if there are different components of connections with the same combination's part code, this is an inconsistency and must be prevented by the exporting system.
⇒ Schematron rule to be added.
- PDM Systems would differentiate e.g. Screw-Nut-Combinations according to the distance between screw head and nut. Since xMCF does not cover this kind of distance, as more general geometric parameters too, tools cannot detect this situation.
- To-Dos: Extend XSD. Extend the Schema. Provide example file(s): With & without template. Formulate a text fragment for the spec.

Add this to the χMCF Specification

Not discussed
2026-10-02.

Label is optional,
but highly recommended,
especially in cases,
when different tools or
partners are involved.

2026-09-15: An according
[issue #113](#) was created.

9.1.1.1 General

The possible XML attributes of the <connection_0d/> element, which describes a point connection (0D connection), are specified in Table 32:

Table 32 — Attributes of element <connection_0d/>

Attributes	Type	Use	Constraint
label	Alphanumeric	Optional	-
ident	Integer	Optional	positive, unique within a χMCF file
<u>quality_control</u>	Alphanumeric	Optional	See clause 8.4.

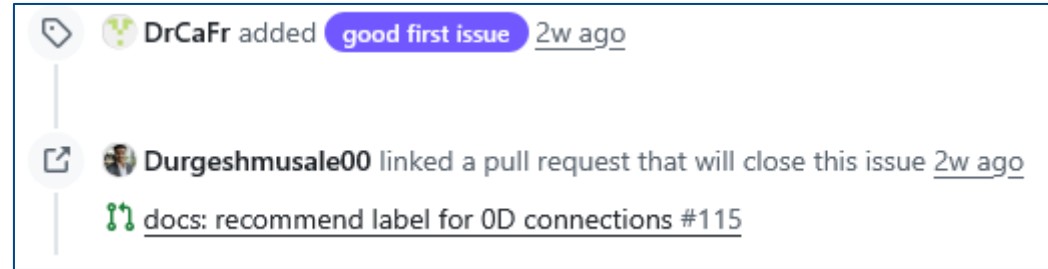
9.1.1.2 Attribute “label”

The label defines the human readable identification of a connection. It can contain a description of the connection or simply an index as an integer.



Who is / knows “Durgeshmusale00”?

Looking at [issue #113](#) , one encounters:

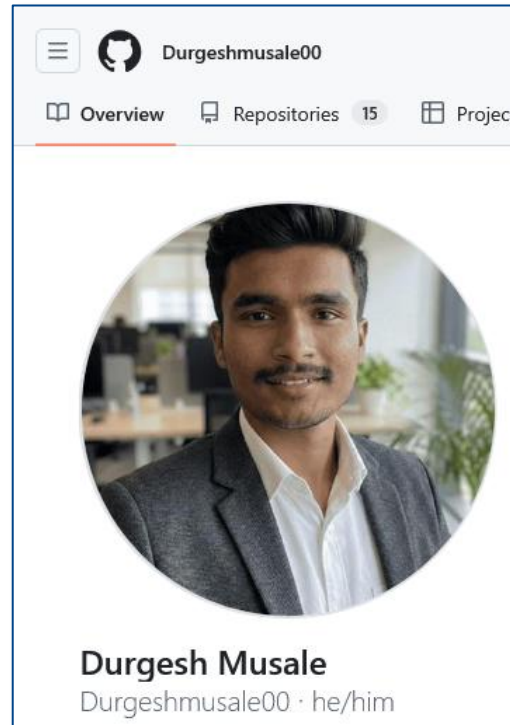


There is a [profile at GitHub](#):

CF to contact him
via LinkedIn.

Done 2026-10-02.

No response
until 2026-10-06.



Other (ISO) Standards

resumed from
2025-11-26 /
2025-12-10 /
2026-01-14
Not discussed
2026-10-02.

[#61](#) suggests to analyze existing standards in this context, such as

- [ISO 13584-511:2006 — Industrial automation systems and integration — Parts library — Part 511: Mechanical systems and components for general use — Reference dictionary for fasteners](#)
— a copy was provided to the WG — for internal use, only.
- [ISO 1891:2009 — Fasteners — Terminology](#)
— a copy was provided to the WG — for internal use, only.

A sincere *thank you* to all donors!

- 2026-09-15: An according [issue #114](#) was created for this analysis.
- Do we need / want to consider more standards?

Next meetings — Suggested new appointments

CF on vacation 2026-10-14 to 21 ⇒ No meetings.

- 2026-11-04, 15:00 CET — focus group on [#105](#) / [#61](#)
 - objective: agree on XML suggestions & examples / draft Word file changes
- 2026-11-11, 15:00 CET — focus group on [#105](#) / [#61](#)
 - objective: agree on XML suggestions & examples / draft Word file changes

CF sent / will send out invitations.

